

Interfacing Keyboard

2020-2021

5[A] Show the software keyboard interfacing process through a flow chart. Describe each step for a 4x4 keyboard. (5)

2019-2020

5[B] When you press a key on your computer, you are activating a switch. There are many different ways of making these switches. Describe any four of them. (4)

5[D] What is key debouncing? Why is it needed? (2)

2018-2019

6[B] 2020-2021 (4)

2017-2018

4[C] Explain Software Keyboard Interfacing Procedure with circuit connections and algorithm. (4)

8279 Programmable Keyboard Display Controller

2020-2021

4[A] Draw the architectural diagram of 8279 programmable keyboard display controller. How does 8279 keyboard work? (5)

2018-2019

5[B] Categorize and explain the operational modes of 8279. How many ways the keyboard is interfaced with the CPU? (4)

2017-2018

5[A] 2020-2021 (6)

8255 PPI

2020-2021

6[D] Obtain the control word when the ports of 8255A are to be used in mode 0 with port-A as output port and port B as input port and port C as output port. (2)

2019-2020

4[A] Show the pin configuration of 8255 PPI and separately describe the operation performed by each unit. (5)

6[B] How can we assign input and output port of 8255? (4)

2018-2019

4[E] 2020-2021 (2)

2017-2018

4[A] 2019-2020 (6)

USB

2019-2020

6[C] In USB communication, a single transaction involves the transmission of up to three packets. Mention the names of these packets and describe the specific services each one provides during the communication process. (3)

2018-2019

5[A] How do the USB peripherals communicate with microprocessor? Give the pin-out for USB type C 3.0 (5)

2017-2018

6[C] What are the basic principles of two-wire handshaking method of data transfer? (2)